Biospectra : ISSN: 0973-7057, Vol. 20(2), September, 2025, pp. 121-128 ***Interdisciplinary Science***
An International Biannual Refereed Journal of Life Sciences



ISSN : 0973-7057

Indigenous Tribal Traditional Knowledge (ITTK) a miss out component in Indian Traditional Knowledge system

Anita Chakraborty[a]*, Kunul Kandir[b], Sanjay Kumar Sinha[a] & Rajiv Kumar[c]
[a]University Department of Botany, S. K. M University, Dumka, Jharkhand, India
[b]University Department of Botany, Ranchi University, Ranchi, Jharkhand, India
[c]University Department of English, S. K. M University, Dumka, Jharkhand, India

Received : 17th April, 2025 ; Revised : 18th May, 2025
DOI:-https://doi.org/10.5281/zenodo.18760068

Abstract- Our country India, including our State universities of Jharkhand has been working unbreakably proactive in the adoption and implementation of the transformative initiatives for the Four-Year Undergraduate programme (FYUGP) UGC's revised guideline and NEP, 2020. The transformative initiatives strongly emphasizes on the integration of the Indian Knowledge System (IKS), including Indigenous Traditional Knowledge (ITK) into the education system. However the integration of tribal traditional knowledge system is almost missing from the larger canvas of the Indigenous Traditional Knowledge (ITK) in NEP, 2020. The present manuscript is an effort to review the potential of Indigenous Tribal Traditional Knowledge (ITK) and advocated its adoption in the Indigenous Traditional Knowledge (ITK) under the overarching structure of NEP, 2020 to achieve the ultimate objective to familiarize the students with India's rich cultural, philosophical, scientific and technical heritage, promoting an appreciation for traditional knowledge systems and their application in the modern context.

Keywords: Indigenous Traditional Knowledge (ITK), Tribal, Jharkhand, NEP, 2020, Policy, Education, Students

INTRODUCTION

The Union Cabinet of India, has recently approves National Education Policy (NEP) in July 2020 with an aim to make India as a global knowledge superpower and also to change the Indian education system targeting the policy adaptation from School to the College level.[1]

The goal of the policy pressed upon the holistic development of student at creativity, critical thinking and problem solving skill through the education. It also aims to equip the students with the skill sets to better adapt in technology driven country in its 21st century and centuries to come. It pressed on the concept of equity and inclusion.

*Corresponding author :
Phone : 9711981579
E-mail : anitagcenator@gmail.com

The policy also aimed to achieve the foundational literacy and numeracy. The policy encourages the use of mother/ local language in education to promote the linguistic diversity and also fostering the belongingness amongst the manatee. The key feature of the policy is the new curriculum structure, integration of vocational studies, promote and integrate technology, policy strengthening of the teachers training and also to reduce the exam burden on the students promoting the conceptual learning ecosystem with critical thinking educational environment.[2] The policy strongly emphasizes on the integration of the Indian Knowledge System (IKS), including Indigenous Traditional Knowledge into the education system.[3] However the integration of tribal traditional knowledge system is

almost missing from the larger canvas of the Indigenous Traditional Knowledge (ITK) in NEP, 2020. The present manuscript is an effort to review the potential of Indigenous Tribal Traditional Knowledge and advocated its adoption in the Indigenous Traditional Knowledge (ITK) under the overarching structure of NEP, 2020 to achieve the ultimate objective, to familiarize the students with India's rich cultural, philosophical, scientific and technical heritage, promoting an appreciation for traditional knowledge systems and their application in the modern context.

METHODOLOGY

Source of Data:

1. Primary Data: Field Survey
   a. Observation,
   b. Interview (face to face) with the academicians, local and tribal community as custodian of the traditional knowledge, to explore the Indigenous traditional identity and their significance in their society and livelihood was analyzed.
2. Secondary Data: Review of Literature
   a. Books
   b. Journals
   c. Internet
   d. Other

Semi-structured questionnaire was prepared to get the information regarding the role and application of Indigenous Traditional Knowledge Systems in the Indian Traditional Knowledge System from the community and different stakeholders concerned and secondary data as well as the records were reviewed and further analyzed to validate the primary data collected. The questionnaire was aimed to collect mostly the qualitative information so that data base required to create the quantitative information can be identified and addressed.

RESULT & DISCUSSION

According to the 2011 Census[4], Scheduled Tribes account for 104 million representing 8.6% of the India's population. They are spread throughout the country over the states of Rajasthan, Gujarat, Maharashtra, Madhya Pradesh, Chhattisgarh, Orissa, Jharkhand, West Bengal and the Northeastern States of Mizoram, Meghalaya, Nagaland, Assam, Manipur, Tripura and Arunachal Pradesh.[5] Article 342 of the Constitution of India defines a 'tribe' as "an endogamous group with an ethnic identity; who have retained their traditional cultural identity; they have a distinct language or dialect of their own; they are economically backward and live in seclusion, governed by their own social norms and largely having a self-contained economy." Other definitions for "tribes" also abound. The word 'tribe' is used for a "socially cohesive unit, associated with a territory, the members of which regard themselves as politically autonomous".[6] The term 'primitive tribes' was first used by western anthropologists to represent, "a primary aggregate of peoples living in a primitive or barbarous condition under a headman or chief".[7]

Tribal traditional knowledge system has its roots from the ancient Indian Philosophy, mathematics, linguistics and ecological awareness. Such knowledge is time tested and had potential which can emphasizes the holistic and sustainable development principles.[8] They belong to "endogamous group" i.e. they do not marry outside their tribes. Kinship forms the basis of tribal organization and has the "egalitarian values" as their distinctive feature, i.e. there are no institutionalized inequalities like the caste system or sex-based inequalities.[9] At one hand where the endogamous culture do not allows the amalgamation or mixing of the another genetic pool with them while their egalitarian values hold a very strong moral foundation of equality in their society, value system and gender orientation in society.

Convention on the Biological Diversity (CBD) firmly advocates respecting, preserving and maintaining, the knowledge, innovation and practices of indigenous and local communities. Traditional lifestyle and practices of the tribal communities are found to be relevant for the conservation and promotion of the sustainable practices. It has ample scope to promote the concept of the equitable sharing of resource and promote its sustainability. Many notable indigenous practices holds significant strategies to address my of the present challenges like, global rise in temperature (°C), greenhouse effect and climate change etc. Tribal communities have devised many of their practices to cope, adapt and mitigate these climate allied uncertainties.

The state of Jharkhand is best example where there is a sizable population of the tribal community contributing to its demographic as well as geographical status in the Country. The tribal community of this region possesses a unique cultural heritage which is characterized by their time tested rich tradition, language, customs and deep interaction with the immediate environment. It has been said that both the tribal community and the rich natural

resource of the State has coexisted and co-emerged in the backdrop of changing environment, ecology and economy of the State. Their co-existence and co-dependence however is always a matter of contest and complex phenomenon which requires a deep insight to understand and explain. The origins of Jharkhand's tribal population can be traced back to ancient times, with archaeological evidence suggesting their presence in the region dating back to thousands of years.

Home to more than thirty distinct tribal communities, including the Santhals, Mundas, Oraons, Ho's, and others, Jharkhand presents a rich tapestry of traditions, languages, art forms, and social customs that are deeply interwoven with the natural environment.[10] The cultural diversity and their practice are both rich and complex, embodying a unique blend of tradition, art, language, and social customs. It offers profound insights into sustainable living, communal harmony, and the deep connections between humans and nature. Few examples of the traditional identity, practices and their significance to the life and livelihood of the tribal community in the state of Jharkhand are presented in Table 1. The tribal traditional knowledge and their practices help in shaping of cultural background, religious institution and social structure of many tribal communities, ethnic groups, and racial stock. The application of tribal traditional knowledge is diverse and has been passed down over generations through oral tradition, kinship network, communal connection, social groups, etc. for judicious management of the resources on which their daily sustenance is dependent upon (Fig. 1).[11]



Fig. 1- Application of Tribal Traditional Knowledge

Due to the globalization, commercialization, scientific dilemma, acculturation in the society, climate uncertainties, climate change, land use and land cover change, engagement of different stakeholder for various interest, policy and reform implementation, migration, infiltration of the other community, decline/lack of interest in the practice of traditional knowledge had already impacted the indigenous tribal traditional practices and its knowledge.

The scenario becomes more intensify when deteriorating ecological situation, persistent poverty, social, political and religious tensions and the reduction of biological and cultural diversity present a polycrisis for which new answers are urgently needed.[12]

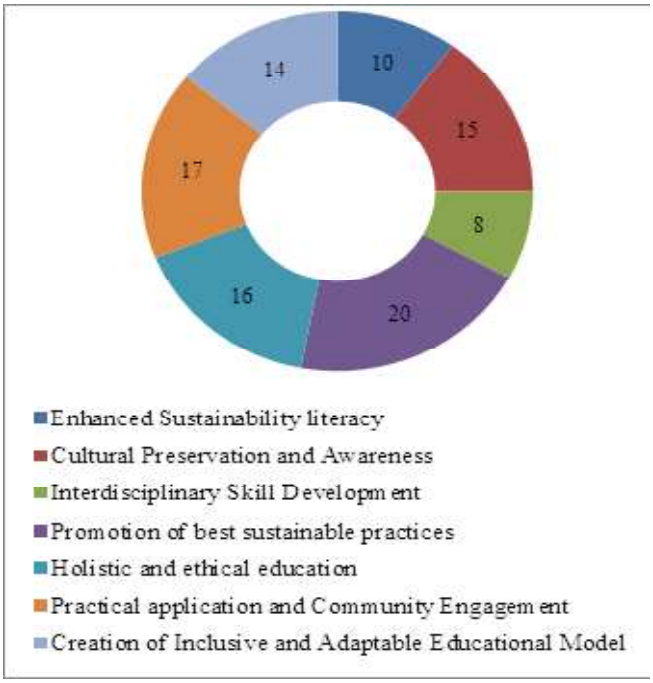


Fig. 2- Role of Tribal Traditional Knowledge

Use of climate smart crop, drought resistant crop varieties, water harvest technology promoting local verities, mixed cropping, agroforestry practices, traditional method of cultivation, wild plants gathering and conservation of the indigenous varieties are some of their adopted strategy or coping mechanism, against the climate change and many technology driven ecology.

Tribal traditional knowledge is the time tested naturally evolved knowledge systems which are strong enough to provide the framework for the conservation of biodiversity and the sustainable management of the resources (Fig. 2). This knowledge has the power of strong adaptation qualities which can mitigate many of the current challenges caused by the industrial development and allied activities. The

knowledge often holds lessons that can be either adopted directly or with minor modification to the existing conservation and sustainability efforts. The principal governing behind the tribal traditional knowledge i.e. efficient, equitable and ecologically can not only be adapted at the local but can be scale up in the national and global level. Adaptation and integration of such principle into the educational and policy framework can help to achieve the millennial sustainable development goal.

Despite the pressures of modernization and globalization, these communities have demonstrated remarkable resilience in preserving their cultural heritage. It is imperative that concerted efforts are made by governmental and non-governmental organizations, scholars, and the communities themselves to document, promote, and sustain these cultural practices. The culture of the tribal people of Jharkhand is a vital part of India's diverse cultural mosaic. Understanding and valuing this heritage not only enriches our knowledge of human civilization but also reinforces the importance of cultural diversity and heritage preservation. Future research and policy initiatives must focus on empowering these communities, ensuring that their rich cultural legacy continues to thrive for generations to come. Integration of the indigenous tribal traditional knowledge will promote its significance and potential in shape the age old practices observed and preserved by them. Their practical application will help to identify and promote the best practices in the current knowledge systems.

This database of knowledge with ancient wisdom requires acceptance, dissemination, and conservation. These sustainable practices should come under the reach and practices across all the other communities and should be transmitted from generation to generation to revive, restore and rediscoverd the best practices across the globe. The incorporation is not only necessary to bridge the gap between the traditional and modern knowledge but can also help in fostering the innovation and creativity.[3] The SWOT analysis of the incorporation of Tribal traditional knowledge system can be presented in Fig. 3

Strength:	**Weakness:**
• Ancient Cultural knowledge, art, culture and craftsmanship • Promote lifestyle education • Traditionally tested and registered in all forms of Vedas • Sustainable and environmentally tested	• Lack of research and available data base • Poor documentation • Language and literature • Translation and interpretation • Curriculum reforms • Lack of dedicated faculty • Lack of policy supports
Opportunities:	**Threats:**
• Multidisciplinary learning environment • Promote cultural identity • Promotion and validation of innovation • Promotion of ethical and sustainable practices • Scope for research collaboration	• Resource creation and allocation • Balancing and incorporating in modern knowledge systems • Cultural Sensitivity and conservativeness • Acceptability and scalability • Resistance to change • Standardization and validation

Fig. 3- SWOT analysis of the incorporation of Tribal traditional knowledge system in IKS

Table 1. An over view of selected traditional identity, practices and their significance to the life and livelihood of the tribal community in the state of Jharkhand (Source: Tiwari S and Phalachandra,B. 2024; Semi structured survey)

Sl. No.	Indigenous traditional Identity		Significance in their life
1	Festivals and Celebrations	Sarhul- The Festival of Flowers, dedicated to the worship of the Sal tree.	Celebrated by the Oraon, Munda, and Ho tribes to marks the advent of the New Year and the arrival of spring in the month of Chaitra (March-April). It holds the spiritual significance and symbolizes the harmony between humans and nature.
		Karma: The Festival of Youth and Nature. The festival involves the worship of the Karam tree, symbolizing prosperity and well-being of the nature.	Celebrated by various tribal communities including the Oraons, Mundas, and Santhals in the month of Bhadra (August-September). Karma is a celebration of youth, fertility, and the mother nature.
		Sohrai: The Harvest Festival	Sohrai, celebrated primarily by the Santhals, is a harvest festival that coincides with the Diwali celebrations in other parts of India. It is held in the month of Karthik (October-November). The festival is marked by the decoration of homes with intricate paintings and murals depicting scenes from tribal mythology and daily life. These paintings are drawn by the women using natural pigments derived from plants and minerals. Cattle, which play a vital role in agriculture, are also worshiped.

		Tusu Parab: The Harvest Celebration and testament to the creative spirit of the tribal.	Tusu Parab is celebrated during the winter months, particularly in the regions inhabited by the Kurmi and Oraon tribes. The festival marks the end of the harvest season and the beginning of the New Year.
		Baha Parab: Celebrated in the months of March each year especially by the santhals tribes	This festival marks the onset of the flowering season ensuring new beginning and productivity.
		Bhagta Parab: The Festival of Strength and Devotion	Bhagta Parab, the festival of devotees, is celebrated mostly by the Oraons and few more tribal communities. It is held in the month of Chaitra. It practices the unique display of physical endurance and devotion.
2	Traditional Music Forms	Folk Songs	Folk songs are an integral part of tribal music in Jharkhand. These are performed during festivals, agricultural activities, and social gatherings. Each tribe has their own distinctive style of folk music, e.g. the Santhal community is known for its "Karam" songs, sung during the Karma festival. Both the lyrics and the songs have repetitive and rhythmic patterns, sung by the use of traditional instruments like the madar (a type of drum) and the bansuri (flute).
		Religious and Ritualistic Songs	Many tribal songs are associated with religious rituals and ceremonies. The Oraons, for example, sing "Sarhul" songs during the Sarhul festival, which marks the beginning of the new year and the onset of spring. These are primarily devotional songs dedicated to the nature, invoking the blessings of deities and spirits.
		Work Songs	Work songs are another significant category of tribal music. These songs are sung collectively while performing agricultural tasks, such as sowing seeds or harvesting crops. The music, lyrics and rhythm of these songs when sung helps to coordinate the efforts of the workers, making their labour more enjoyable and less strenuous.
3	Traditional Dance Forms	Chhau Dance	It is a semi-classical dance form popular amongst the Santhals. This dance form combines the elements of martial arts, acrobatics, and strong mythological storytelling (like the Ramayana-Asur Vadha, various episode of the Mahabharata. Performers of this form of dance, wears an elaborate masks and costumes. The vigorous movements and intricate choreography of Chhau make it a visually captivating performance.
		Santhal Dance	It is performed during Sarhul and Sohrai festivals, as well as during the social gatherings and celebrations. Men and women form circles or semi-circles holding hands together with each other to dance in rhythm using traditional musical instruments The dance steps and movements are simple though highly graceful, reflecting the harmonious relationship between the people and its natural surroundings.
		Paika Dance	The Paika dance, primarily performed by the Munda tribe, is a martial dance that showcases the valor and bravery of the tribal warriors. The dancers, wore the traditional warrior attire, perform vigorous movements that mimic battle scenes. They use shields and swords as props, and the dance is often accompanied by the beating of drums and other percussive instruments.
		Jhumar Dance	Jhumar is a popular folk dance form in Jharkhand. It is performed during harvest festivals and many social events. The dance is characterized by its slow, rhythmic movements and the graceful swaying of the dancers. Jhumar songs, sung during the dance, often narrate the stories of love, nature, and rural life.
		Domkach Dance	Domkach is a community dance performed by the Oraons and other tribes during weddings and other joyous occasions. Men and women dance together in a circle, holding hands and swaying to the rhythm of folk songs and music. The dance symbolizes unity and collective celebration.
4	Musical Instruments	Madal/Mandar	It is a double-headed drum, played with both the hands. It is a key instrument in many tribal music performances.
		Bansuri/Banshi	It is a bamboo flute that produces a melodious sound. It is often used in accompaniment with songs and dances
		Nagara	It is a large drum played with sticks, used during festivals and ceremonial occasions
		Shehnai	It is a wind instrument that adds a distinct sound to the music, often played during weddings and religious ceremonies

5	Cuisine	Dhuska	Dhuska is a traditional deep-fried dish made from a batter of rice and lentils. It is usually served with gram-potato/vegetable curry or chutney. This dish is commonly prepared during festivals and special occasions.
		Handia	Handia is a traditional rice beer made by fermenting rice with herbal starters known as "ranu/bakhar tablets." It is a popular beverage among the tribal communities and is consumed during social gatherings and festivals.
		Pitha	Pitha refers to a variety of rice flour dumplings that can be either savory or sweet. They are steamed or fried and often stuffed with ingredients like lentils, jaggery, or coconut or even dal.
		Sandhna Saag	This dish is made from tender bamboo shoots, which are either stir-fried or cooked with spices. Bamboo shoots are a seasonal delicacy and are highly valued for their unique taste and nutritional benefits.
		Sanai Phool	This dish is prepared from the flowers of the sanai (hemp) plant. It is typically cooked with mustard oil, garlic, and chilies, offering a distinct flavor and rich nutritional content.
		Rugra:	Rugra is a type of wild mushroom that is highly prized among the tribal communities. It is usually cooked with spices to make a rich, flavorful curry.
		Chhilka Roti	Chilka roti is a type of pancake made from a mixture of rice flour and gram flour. It is often served with chutney or vegetable curry.
6	Traditional Dress	Men's Attire-bhabla	Tribal men typically wear a dhoti or loincloth, known locally by various names such as "lungee" or "bhabla." These garments are usually made from cotton and are draped around the waist, extending up to the knees or ankles.
		Women's Attire-panchhi/parhan	Tribal women in Jharkhand wear sarees or similar garments known as "panchi" and "parhan." These sarees are generally made from cotton and are often dyed in bright colors with simple yet striking patterns. The saree is usually draped in a unique style which allows ease of movement during daily house chores and many festivals.
		Special Occasions	During festivals, weddings, and other special events, both men and women dress in more elaborate and colorful attire. Women may wear sarees with intricate designs and borders, while men might do more ornate kurta or an angarkha (a traditional upper garment).
7	Ornaments	Necklaces	Necklaces are a common adornment among tribal women. These necklaces are often made from beads, coins, and shells. One of the most popular types is the "Hasauli," a heavy necklace made from silver or other metals, often worn during special occasions.
		Earrings	Earrings are widely worn by both men and women in some tribal communities. They range from simple studs to elaborate danglers made from metals, beads, and stones. The "Chik Chiki" is a traditional earring design popular among the Santhal women, characterized by its circular shape and intricate detailing in the making of the ear piece.
		Bangles and Bracelets	Bangles made from various materials such as glass, metal, and lac are commonly worn by tribal women. These bangles are often colorful and worn in multiples. Men sometimes wear bracelets made from metal or beads, which can have symbolic meanings related to their tribe or personal achievements.
		Anklets and Toe Rings	Anklets, known as "Payal" or "Paijan" are worn by women, especially during festivals and dances. These anklets are usually made of silver and can be simple or adorned with small bells. Toe rings are also worn by married women as a symbol of their marital status.

CONCLUSION

Integrating tribal traditional knowledge systems in traditional Knowledge System into higher education system, as encouraged by NEP 2020, will be one of the judicious innovative approaches towards creating an education system that is deeply rooted in cultural heritage of every known corner of our nation with its diverse set of communities. It will give a platform for all the innovative, time tested and sustainable practices which not only have to potential to serve where it has originated but can serve the larger set of people if need and adopted (Fig.3). This noble approach will not only provide the due respect to the Indian culture but will also promote the Intellectual identity of the tribal customary to the rest of the world. Furthermore it will provide a self-esteemed ecosystem for all trial communities for wider acceptance and adaptation. The principle of multidisciplinary approach in NEP, 2020 in the new education system will also be achieved. It will promote

both the innovation and traditional wisdom of this country. The incorporation of tribal traditional knowledge into the Indian Traditional Knowledge has high potential in creating culturally rich and ethically validated system of education for both the learners and the educator in the present generation and years to come.

REFERENCE

1. **Public Information Bureau, 2022.** https://www.pib.gov.in/indexd.aspx.
2. **MHRD, 2020.** https://www.education.gov.in/sites/upload_files/mhrd/files/NEP_ Final_English_0.pdf
3. **Mishra, S, Tripathy, A. B., Patro P. R. 2024.** Integrating Traditional Indian Knowledge System in Indian Higher Education) (In Nep 2020 Perspectives). *International Journal of Current Science.* **14 (3):**346-352.
4. **https://www.census2011.co.in/census/district/97-dumka.html**.
5. **United Nations Development Programme, 2008.** Paharias the Struggle of a tribe for Recognition. 1-40 pp.
6. **Mitchell, J. A. 1994.** A World without Tribes? Tribal Rights of Self-Government and the Enforcement of State Court Orders in Indian Country. *The University of Chicago Law Review.* **61(2):** 707-732.
7. **Sills, D.L. 1976.** International Encyclopedia of Social Sciences. **16:** 147-150. New York: MacMillan and Free Press.
8. **Lal, K. S, Srivastava, S, Narayan V, Pal N, Kumar R and Sinha S. 2024.** Indian Knowledge System Challenges and its application in Higher Education for Sustainable Future Development. *Library Progress International*. **44(3):** 11931-11941.
9. **Shukla Y. and Shahdeo, GSN Lal. 2019.** Struggle for Existence: Korwa Tribes of Jharkhand. *JETIR*. **6(6):** 461-465.
10. **Tiwari S. & Phalachandra B. 2024.** Culture of Tribal People of Jharkhand: An Overview. *International Journal of Research Publication and Reviews*. 5(6): 150-159.
11. **https://ebooks.inflibnet.ac.in/antp05/chapter/indigenous-knowledge-and-sustainability/**
12. **Raul M., Majumdar K. and Chatterjee D., 2021.** Indigenous Knowledge and Endogenous Development: Exploring Survival Strategy of a Tribal Community of Jharkhand, India. *International Journal of Research and Review (ijrrjournal.com).* **8(8):**156-170.

***